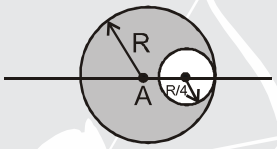
SYLLABUS : CENTER OF MASS

1. The centre of mass of a non uniform rod of length L whose mass per unit length varies as $\rho = kx^2/L$ (where k is a constant and x is the distance measured from one end) is at the following distance from the same end.

(A) 3L/4 (B) L/4 (C) 2L/3 (D) L/3

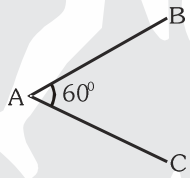
2. The centre of mass of the shaded portion of the disc is : (The mass is uniformly distributed in the shaded portion) :



(A) $\frac{R}{20}$ to the left of A (B) $\frac{R}{12}$ to the left of A

(C) $\frac{R}{20}$ to the right of A (D) $\frac{R}{12}$ to the right of A

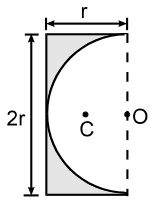
3. A uniform wire of length ℓ is bent into the shape of 'V' as shown. The distance of its centre of mass from the vertex A is



(A) $\ell/2$ (B) $\frac{\ell\sqrt{3}}{4}$ (C) $\frac{\ell\sqrt{3}}{8}$ (D) None of these

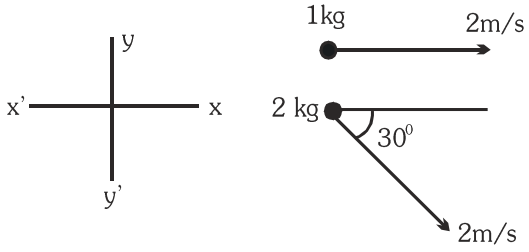
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4. A semicircular portion of radius 'r' is cut from a uniform rectangular plate as shown in figure. The distance of centre of mass 'C' of remaining plate, from point 'O' is :



(A) $\frac{2r}{(3-\pi)}$ (B) $\frac{3r}{2(4-\pi)}$ (C) $\frac{2r}{(4+\pi)}$ (D) $\frac{2r}{3(4-\pi)}$

5. The velocity of centre of mass of the system as shown in the figure



(A) $\left(\frac{2-2\sqrt{3}}{3}\right)\hat{i}-\frac{1}{3}\hat{j}$ (B) $\left(\frac{2+2\sqrt{3}}{3}\right)\hat{i}-\frac{2}{3}\hat{j}$ (C) $4\hat{i}$ (D) None of these

6. Two particles of mass 1 kg and 0.5 kg are moving in the same direction with speed of 2m/s and 6m/s respectively on a smooth horizontal surface. The speed of centre of mass of the system is:

(A) $\frac{10}{3}$ m/s (B) $\frac{10}{7}$ m/s (C) $\frac{11}{2}$ m/s (D) $\frac{12}{3}$ m/s

7. Two particles having mass ratio n : 1 are interconnected by a light inextensible string that passes over a smooth pulley. If the system is released, then the acceleration of the centre of mass of the system is :

(A) $(n-1)^2 g$ (B) $\left(\frac{n+1}{n-1}\right)^2 g$ (C) $\left(\frac{n-1}{n+1}\right)^2 g$ (D) $\left(\frac{n+1}{n-1}\right) g$

8. There are two particles of same mass. If one of the particles is at rest always and the other has an acceleration $\vec{a}$. Acceleration of centre of mass is

(A) zero (B) $\frac{1}{2}\vec{a}$ (C) $\vec{a}$

(D) centre of mass for such a sxystem can not be defined.

9. A particle of mass m is made to move with uniform speed v_0 along the perimeter of a regular hexagon, inscribed in a circle of radius R. The magnitude of impulse applied at each corner of the hexagon is

(A) $2mv_0\sin\pi/6$ (B) $mv_0\sin\pi/6$ (C)$mv_0\sin\pi/3$ (D) $2mv_0\sin\pi/3$

10. A particle of mass 4m which is at rest explodes into three fragments. Two of the fragments each of mass m are found to move with a speed 'v' each in mutually perpendicular directions. The minimum energy released in the process of explosion is :

(A) $(2/3)\,mv^2$ (B) $(3/2)\,mv^2$ (C) $(4/3)\,mv^2$ (D) $(3/4)\,mv^2$

11. A man of mass 'm' climbs on a rope of length L suspended below a balloon of mass M. The balloon is stationary with respect to ground. If the man begins to climb up the rope at a speed v_{rel} (relative to rope). In what direction and with what speed (relative to ground) will the balloon move?

(A) downwards, $\frac{mv_{rel}}{m+M}$ (B) upwards, $\frac{Mv_{rel}}{m+M}$

(C) downwards, $\frac{mv_{rel}}{M}$ (D) downwards, $\frac{(M+m)v_{rel}}{M}$

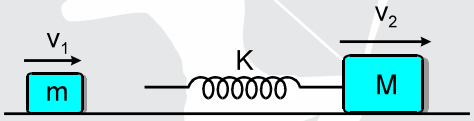
12. A particle of mass 4m which is at rest explodes into masses m, m & 2m. Two of the fragments of masses m and 2m are found to move with equal speeds v each in opposite directions. The total mechanical energy released in the process of explosion is

(A) mv^2 (B) $2mv^2$ (C) $1/2\ mv^2$ (D) $4mv^2$

13. Two masses of 1g and 4g are moving with equal kinetic energy. The ratio of the magnitude of their linear momentum is -

(A) 1 : 1 (B) 1 : 2 (C) 1 : 3 (D) 1 : 4

14. Two blocks of masses m and M are moving with speeds v_1 and v_2 ($v_1 > v_2$) in the same direction on the frictionless surface respectively, M being ahead of m. An ideal spring of force constant k is attached to the backside of M (as shown). The maximum compression of the spring when the block collides is :



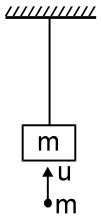
(A) $v_1\sqrt{\frac{m}{k}}$ (B) $v_2\sqrt{\frac{M}{k}}$

(C) $(v_1 - v_2)\sqrt{\frac{mM}{(M+m)K}}$ (D) None of above is correct.

15. A ball of mass 50 gm is dropped from a height h = 10 m. It rebounds losing 75 percent of its kinetic energy. If it remains in contact with the ground for Δt = 0.01 sec., the impulse of the impact force is : (take g = 10 m/s²)

(A) 1.3 N–s (B) 1.06 N-s (C) 1300 N–s (D) 105 N–s

16. A bullet of mass m moving vertically upwards instantaneously with a velocity 'u' hits the hanging block of mass 'm' and gets embedded in it, as shown in the figure. The height through which the block rises after the collision. (assume sufficient space above block) is :



(A) $u^2/2g$ (B) u^2/g (C) $u^2/8g$ (D) $u^2/4g$

17. A bullet of mass m = 50 gm strikes ($\Delta t \approx 0$)a sand bag of mass M = 5 kg hanging from a fixed point, with a horizontal velocity $\vec{v}_p$. If bullet sticks to the sand bag then just after collision the ratio of final & initial kinetic energy of the bullet is:

(A) 10^{-2} (B) 10^{-3} (C) 10^{-6} (D) 10^{-4}

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18. A solid iron ball A of radius r collides head on with another stationary solid iron ball B of radius 2r. The ratio of their speeds just after the collision (e = 0.5) is :

(A) 3 (B) 4 (C) 2 (D) 1

19. A body of mass 1 kg strikes elastically with another body at rest and continues to move in the same direction with one fourth of the initial velocity. The mass of the other body is –

(A) 0.6 kg (B) 2.4 kg (C) 3 kg (D) 4 kg

20. A super-ball is to bounce elastically back and forth between two rigid walls at a distance d from each other. Neglecting gravity and assuming the velocity of super-ball to be v_0 horizontally, the average force (in large time interval) being exerted by the super-ball on one wall is :

(A) $\frac{1}{2}\frac{mv_0^2}{d}$ (B) $\frac{mv_0^2}{d}$ (C) $\frac{2mv_0^2}{d}$ (D) $\frac{4mv_0^2}{d}$

21. A sphere of mass m moving with a constant velocity hits another stationary sphere of the same mass. If e is the coefficient of restitution, then ratio of speed of the first sphere to the speed of the second sphere after head on collision will be :

(A) $\left(\frac{1-e}{1+e}\right)$ (B) $\left(\frac{1+e}{1-e}\right)$ (C) $\left(\frac{e+1}{e-1}\right)$ (D) $\left(\frac{e-1}{e+1}\right)$

22. Ball 1 collides head on with an another identical ball 2 at rest. Velocity of ball 2 after collision becomes two times to that of ball 1 after collision. The coefficient of restitution between the two balls is :

(A) e = 1/3 (B) e = 1/2 (C) e = 1/4 (D) e = 2/3

23. A particle moving horizontally collides with a fixed plane inclined at 60° to the horizontal. If it bounces vertically, the coefficient of restitution is:

(A) $\frac{1}{\sqrt{3}}$ (B) $\frac{2}{\sqrt{3}}$ (C) $\frac{1}{3}$ (D) None of these

24. If the thrust force on a rocket which is ejecting gases with a relative velocity of 300 m/s, is 210 N. Then the rate of combustion of the fuel will be :

(A) 10.7 kg/sec (B) 0.07 kg/sec (C) 1.4 kg/sec (D) 0.7 kg/sec

25. A ball moves horizontally in a closed box making several collisions with the walls. The box is kept on a smooth horizontal surface. The velocity of the centre of mass during the motion of the ball.

(A) of the box remains constant (B) of the box plus the ball system remains constant

(C) depends on value of e (D) of the ball relative to the box remains constant

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ANSWER KEY

1.	(A)	**2.**	(A)	**3.**	(C)	**4.**	(D)	**5.**	(B)
6.	(A)	**7.**	(C)	**8.**	(B)	**9.**	(A)	**10.**	(B)
11.	(A)	**12.**	(B)	**13.**	(B)	**14.**	(C)	**15.**	(B)
16.	(C)	**17.**	(D)	**18.**	(C)	**19.**	(A)	**20.**	(B)
21.	(A)	**22.**	(A)	**23.**	(C)	**24.**	(D)	**25.**	(B)